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IDENTIFYING ATTRIBUTES OF CANCER TREATMENTS: WHAT DO STAKEHOLDERS CONSIDER IMPORTANT?

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OBJECTIVES: Increasing cancer therapy costs have produced a recent emergence of numerous value frameworks in oncology. However, variation within frameworks and lack of clarity regarding inclusion criteria for value 'attributes' presents a need for further research. This study aims to identify such attributes based on the priorities of different stakeholder groups in order to develop a new conceptual framework for value in oncology. **METHODS:** January-February 2017 three focus groups (cancer patients, oncology physicians and nurses) were conducted in New York, USA. Using nominal group technique (NGT), groups were tasked to identify and prioritise cancer therapy attributes. Qualitative thematic analysis of transcripts was conducted. **RESULTS:** 30 attributes were identified. The focus groups highlighted that beyond a primary consideration of the health gains of the therapy (efficacy and toxicity), priorities varied. Long-term adverse effects (sequelae), alternative treatment options, quality of evidence, how well established the treatment is and reputation of the treating oncologist/centre were prioritized by patients (n=8) whilst the nurses (n=10) preferences focussed on mode of administration, quality of life, communication and treatment innovation. The physicians (n=6) prioritised the burden and inconvenience of treatments (to patients and carers), functional outcomes, the financial toxicity to patients, and the societal costs and consequences of the treatment with reference to the disease burden to be addressed. From the thematic analysis a conceptual framework was developed whereby attributes where assigned across health-related, cost-related and non-health-related categories. **CONCLUSIONS:** Whilst cancer therapy health-gains are prioritised, value frameworks should include considerations beyond health gains and potentially incorporate differences across stakeholders. The study outlines that the relative importance of attributes varies across stakeholder groups. Results of this study are currently informing the development of a discrete choice experiment (DCE) to elicit specific weightings of preferences across stakeholders.

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EVALUATING ROBOTREVIEWER FOR AUTOMATED RISK OF BIAS ASSESSMENT IN A SYSTEMATIC REVIEW: A CASE STUDY

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OBJECTIVES: Risk of bias (RoB) assessment is an important part of a systematic review and hence the production of health technology assessment. However, it is a time consuming, subjective and labour intensive process, and disagreements on a study's RoB between reviewers are common. In response, novel software tools have emerged which aim to support this process. RobotReviewer is a free web-based machine learning system that aims to automate RoB assessments of randomised controlled trials (RCTs). RobotReviewer has been tested internally by its developers where it performed well, but to date we have not identified any published independent evaluations. We compared and evaluated RobotReviewer against the current standard for RoB assessment, defined as double, independent, human researcher assessment with disagreements resolved by a third reviewer. **METHODS:** A case study has been undertaken where RobotReviewer was tested on a subset of RCT papers that had been previously assessed by two independent human researchers, as part of a systematic review. The results of the automated (i.e. RobotReviewer) assessments were compared with the manual, human reviewer assessments for similarity at each RoB domain in accordance with the Cochrane Risk of Bias Tool. **RESULTS:** 35 papers reporting an RCT were assessed for RoB by two independent human reviewers and RobotReviewer. The results of manual (i.e. human reviewer) and automated (i.e. RobotReviewer) RoB assessment were compared. The mean level of agreement between RobotReviewer and double, independent, human researcher assessment was 72% across all papers. 25% agreement was achieved across three papers, 50% agreement was achieved across eight papers, 75% agreement was achieved across 14 papers, and 100% agreement was observed across 10 papers. **CONCLUSIONS:** This study has provided practical insight into the current effectiveness of employing a machine learning system to support RoB assessment in a systematic review, as part of a health technology assessment.

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AN EXPLORATION OF THE METHODOLOGY OF PUBLISHED SYSTEMATIC LITERATURE REVIEWS ON "BURDEN OF DISEASE"

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OBJECTIVES: Understanding the burden of diseases is important for policymakers and health technology assessment bodies. However, there is no established methodological guidance on systematically identifying evidence relating to the clinical, economic or humanistic burden of a disease. We identified systematic literature reviews (SLRs) on burden of disease and compared their methodologies to look for common themes or key differences. **METHODS:** We searched MEDLINE via PubMed for articles with "burden" in the title, and limited the results using the "Systematic Reviews" filter. Articles were screened in two stages by one reviewer, with input from a second reviewer when uncertain. Eligible articles had to report sufficient details on ≥2 methodological aspects: evidence sources, database search terms, eligibility criteria, or quality assessment tools. **RESULTS:** From 834 records identified through PubMed, 251 were SLRs on disease burden. Over 80% of SLRs reported clinical burden (e.g. prevalence, mortality, morbidity), 50% reported economic burden (e.g. costs, resource use, absenteeism, productivity losses), 20% reported humanistic burden (e.g. quality of life and other patient-reported outcomes) and 10% reported caregiver burden. Methodology was highly heterogeneous, partly reflecting diverse definitions of "burden" used across SLRs. Almost all SLRs searched MEDLINE and Embase, in addition to 0–20 other sources of evidence (e.g. other databases, congress abstract books, reference lists). Database search terms did not generally use validated filters for study designs; many did not use any specific search terms for burden outcomes. Formal quality assessment was only performed in half of the included studies, using a variety of published checklists. **CONCLUSIONS:** The variation in approaches for conducting SLRs on burden of disease may make it difficult for policymakers to compare the results of SLRs in different disease areas. Consideration of the similarities and differences between the methodologies identified in this review will be useful to inform future SLRs.

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3-STEPS STRAGETY FOR PATIENTS EARLY ACCESS TO AN UNAUTHORIZED MEDICATION / DEVICE PROJECT DESIGN (BASED ON 23 PROJECTS ANALYSES INCLUDING EAP, CUP, NPP, IND, ATU)

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OBJECTIVES: Every expanded access project is unique, thus the goal of this research was to develop a systemic approach that could be implemented for any project on early access independent of the design. **METHODS:** 3-steps strategy has been developed, verified and tested on 23 expanded access projects of various designs, including EAP, CUP, NPP, treatment IND, and ATU. The first step in the strategy is to complete the objectives matrix: primary and secondary. Using market authorization plan, to mark the lists of countries as 'proactively go', 'reactively go' and 'no go'. Based on the matrix for the objectives (ethical – to treat patients only; to get early real-word data/long term FUP, reimburse the cost/launch best price policy) the strategy at a second step creates an algorithm of the best submission approach per every country/site/patient or its combinations using the dedicated regulatory database. During the third step the strategy reflects a matrix of the objectives and submission processes and implements accordingly into the dedicated project processes, plans and systems unique per country/site/patient. **RESULTS:** 3-steps strategy supports multiple designs of the project for a successful early access but also downstream commercialization following marketing approval. For example, in countries where charging is permitted, prices used in early access will anchor payer expectations around price points. Also, familiarity with drug usage in early access should enable smoother uptake upon launch. 3-steps strategy was found to be cost-effective, supporting the timelines for launch, and fulfilling the objectives of the client and other involved stakeholders. **CONCLUSIONS:** Despite the complexity and variety of early access project designs in worldwide arena, 3-steps strategy can be implemented in each scenario. It results in fewer burdens for all parties involved, and is effective in meeting the unmet requirements of patient population to be treated within the speed of EAP.

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EMBEDDING RANDOMIZED CONTROLLED TRIALS INTO MEDICAL PRACTICE

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OBJECTIVES: To provide insight in how Pragmatic Clinical Trial (PCT) features can be embedded in RCTs and especially how registries and other routinely collected health data can be used to reduce costs, increase speed of recruitment and limit the amount of additional work needed for investigators, while keeping sufficient quality and acceptability. **METHODS:** Literature review and assessment of ongoing trials. **RESULTS:** In the TASTE and VALIDATE-SWEDEHEART study, among others, electronic databases were present, so that it was possible to define and fine-tune inclusion and exclusion criteria based on medical practice before the start of the trial as well as to test the relevance of CRFs and to set expectations around drop-out, missing values and site recruitment. For the Validate-SWEDEHEART trial, data substitute and support/validate RCT data and support its generalizability, using hybrid systems where adjudication is possible. Long term follow up and extension studies were enabled as most patients stay in the electronic databases. This shows that traditional RCTs can be made more pragmatic regarding follow up, eligibility criteria, recruitment, organization, setting and primary outcomes, while adherence and treatment prescription needs to be strictly defined to assign the relative outcome to treatment. Mitigation of potential risks regarding blinding, among others, will be discussed. **CONCLUSIONS:** RCTs can be embedded in current medical practice in a variety of ways to reduce costs and improve pace of study execution while remaining acceptable.

RESEARCH ON METHODS – Conceptual Papers

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GAMIFICATION, WHAT IS IT AND HOW CAN IT BE USED IN HEALTH OUTCOMES RESEARCH?

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BACKGROUND: Gamification is a novel approach which aims at applying elements from game design to other contexts. This method seeks to improve user experience and increase engagement, with medium to long-term benefits on behavior. So far gamification has been successfully applied to Business, e-Learning and Marketing, and starts attracting more and more attention in e-Health. **METHODS:** The aim of this paper is to introduce theoretical aspects on gamification and its possible impact on wellness and health-related context. By providing real-world example of how it is currently applied in health, we will showcase the means and methods that can increase patients and health professionals' fun, engagement, and communication. Motivation plays a key role when engaging into a brand new activity in autonomy for a recurring amount of time (intrinsic motivation vs extrinsic motivation). There is a wide variety of elements related to motivation affordances (e.g., points, leaderboards, feedback), however to be effective they must address the social situation of gameplay as well as the specificity of each health context. Platform of